Predicting Smartphone Obsolescence Time Using Hardware and Market Features

Project Report
Professional Diploma in Artificial Intelligence

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Contents

Acknowledgement **iii**

Abstract **iv**

1 Introduction **1**

2 Proposed Methodology **3**
2.1 Dataset Description 3
2.2 Model Selection 5
2.2.1 Regression Models 5
2.2.2 Classification Models 7
2.3 Gradio Interface Implementation 8

3 Implementation **10**

4 Implementation **10**
4.1 Dataset Creation 10
4.2 Data Preprocessing 10
4.3 Regression Implementation 11
4.4 Classification Implementation 11
4.5 Gradio Interface Implementation 12

5 Result Analysis **14**
5.1 Correlation 14
5.2 Regression Results 14
5.3 Classification Results 16
5.4 Gradio Interface 19

6 Conclusion **21**

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Abstract

Smartphones undergo rapid technological aging as new hardware capabilities, software features, and market expectations evolve each year. Predicting when a smartphone becomes outdated is essential for guiding consumer choices, improving device design, and reducing electronic waste. This project presents a machine learning–based approach to estimate smartphone obsolescence time using a combination of hardware specifications, software support indicators, and market-driven attributes. A comprehensive synthetic dataset containing 80,000 smartphone entries was generated to simulate real-world diversity in device configurations. After preprocessing through encoding, normalization, and correlation analysis, both regression and classification models were implemented to predict continuous obsolescence time and to categorize devices into longevity classes such as Best, Good, Below Average, and Worst. Evaluation metrics such as Mean Squared Error (MSE), R^2 Score, Accuracy, and the Confusion Matrix were used to assess the models. Results showed that Multiple Linear Regression provided the most reliable performance for regression tasks, while Support Vector Machine (SVM) achieved the highest accuracy in classification. To enhance accessibility, a Gradio interface was developed, enabling users to input smartphone features and obtain real-time predictions. The study demonstrates the effectiveness of machine learning in modeling device aging, offering valuable insights for consumers, manufacturers, and sustainability research.

1 Introduction

The rapid evolution of mobile technology has transformed smartphones into indispensable tools for communication, productivity, and entertainment. However, this rapid technological advancement also leads to a significant challenge: smartphones tend to become outdated within a short span of time. As newer models with improved hardware capabilities, enhanced software support, and upgraded features are introduced, older devices gradually lose their relevance, performance efficiency, and long-term usability. This phenomenon, often referred to as smartphone obsolescence, directly affects consumer experience, purchasing decisions, and electronic waste generation. Understanding and predicting the point at which a smartphone becomes outdated is therefore crucial in both consumer and industry contexts. It allows users to make more informed purchasing decisions, manufacturers to design devices with longer effective lifespans, and researchers to study the environmental impact of device turnover.

This project focuses on developing a machine learning–based predictive system to estimate the obsolescence time of a smartphone using its hardware characteristics, software support details, and relevant market-driven features. Instead of relying on a single dimension such as hardware degradation or resale value, the study combines multiple factors—including processor speed, RAM capacity, battery life, display technology, connectivity standards, camera performance, brand strength, update support duration, and pricing trends—to create a more holistic representation of device longevity. To achieve this, a large synthetic dataset consisting of 80,000 smartphone entries was generated, simulating real-world diversity in device specifications and market conditions. This dataset captures realistic patterns observed in the smartphone industry over the past decade, allowing the models to learn how different features contribute to short-term and long-term device performance.

The implementation of this project follows a structured methodology. First, the synthetic dataset was created using probabilistic distributions to ensure realistic variation in smartphone specifications. Next, preprocessing techniques such as label encoding, feature scaling, and correlation analysis were applied to prepare the data for machine learning. Once the dataset was ready, both regression and classification models were developed to address two prediction objectives: estimating the obsolescence time in years and categorizing the device into longevity classes such as Best, Good, Below Average, and Worst. Various algorithms were trained and evaluated, including Multiple Linear Regression, Support Vector Regression, k-Nearest Neighbors, Decision Trees, Random Forests, Naive Bayes, and Support Vector Machine (SVM). To assess model performance, appropriate evaluation metrics—such as Mean Squared Error (MSE) and R^2 Score for regression, and Accuracy and Confusion Matrix for classification—were used to determine the reliability and effectiveness of each model.

Among the models evaluated, Multiple Linear Regression emerged as the best-performing regression

method, demonstrating that the relationship between smartphone features and longevity can be effectively captured by a simple linear structure. For classification, Support Vector Machine (SVM) achieved the highest accuracy, proving its capability to construct well-defined decision boundaries even in the presence of overlapping feature distributions. To enhance the practical usability of the developed system, a Gradio-based user interface was implemented. This interface allows users to input smartphone specifications interactively and receive real-time predictions regarding the device's obsolescence category, thereby transforming the analytical model into a functional tool suitable for both academic and real-world usage.

Overall, this project demonstrates the practical application of machine learning in predicting smartphone aging and relevance. It highlights how data-driven techniques can provide insights into device longevity and assist in sustainable decision-making. The study lays a strong foundation for future work, such as using real-world datasets, modelling usage behavior over time, incorporating software aging, or optimizing device design for longer lifespan. By bridging technological understanding with predictive analytics, this project contributes meaningfully to the fields of consumer electronics research, lifecycle analysis, and sustainable technology development.

2 Proposed Methodology

2.1 Dataset Description

The dataset used in this project is a synthetically generated dataset created using Python libraries such as NumPy and Pandas. The objective behind generating a synthetic dataset was to simulate a large, diverse, and realistic collection of smartphone specifications that reflect actual trends in the mobile industry. Since obtaining a real-world dataset with complete hardware, software support, and market attributes is challenging, this synthetic dataset allows controlled modelling of obsolescence patterns while maintaining statistical consistency and variability.

The dataset contains 80,000 smartphone entries and a total of 17 attributes, of which 15 are input features describing the device, and 2 are target variables representing the predicted obsolescence indicators. All feature values were generated within realistic ranges based on current smartphone markets (2015–2025), ensuring that the dataset closely resembles real-world device distributions.

Below is a detailed explanation of each feature:

- Processor Speed (GHz) - Represents the clock speed of the smartphone's CPU. Higher clock speeds indicate better performance and typically contribute to improved device longevity.
- RAM Size (GB) - Measures the memory capacity available for multitasking. Devices with larger RAM experience slower performance degradation and stay relevant longer.
- Storage Capacity (GB) - Indicates internal storage space. Limited storage leads to faster slowdown and earlier obsolescence, whereas higher storage offers better long-term usability.
- Battery Capacity (mAh) - Denotes the energy capacity of the battery. Higher battery capacities reduce charge cycles and slow down battery degradation, contributing to extended device lifespan.
- Display Technology (Categorical: LCD, AMOLED, OLED) - Describes the type of display panel used. OLED and AMOLED technologies are more modern and power-efficient compared to LCD, offering better longevity.
- Connectivity (Categorical: 4G, 5G) - Represents network capability. 5G smartphones tend to remain relevant longer due to future-proof network compatibility.
- Camera Score (1–10) - A composite rating indicating overall camera performance. Although not directly related to hardware aging, it influences market value and perceived longevity.

- Update Support Years - The number of years the manufacturer provides major OS updates. Longer support periods significantly extend the functional lifespan of the device.
- Security Updates per Year - Indicates the frequency of security patch releases. Devices with frequent security updates remain more secure and usable for extended periods.
- Brand Score (1–10) - Represents an estimated measure of brand reliability, customer trust, and long-term support. Higher brand scores often correlate with longer-lasting devices.
- Release Year (2015–2025) - The year the device model was launched. Newer models generally incorporate advanced technology and receive longer support.
- Initial Price (USD) - The launch price of the device. Premium models usually contain higher-quality components and receive extended software support, resulting in slower obsolescence.
- Resale Value (USD) - Estimated secondary market value. This reflects depreciation and gives insight into the remaining utility of the device.
- Customer Rating (2.5–5.0) - Aggregated user satisfaction rating. Higher ratings imply better performance durability and consumer approval.
- Depreciation Rate (%) - Annual value loss of the device. Higher depreciation suggests quicker market obsolescence.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Processor_Speed_GHz	RAM_GB	Storage_GB	Battery_mAh	Display_Tech	Connectivity	Camera_Score	date_Support_Ye	rity_Updates_Per_	Brand_Score	Release_Year	itial_Price_US	sale_Value_U	Customer_Rating	ciation_Ra	Obsolescence_Time_Years	Obsolescence_Category	
2	2.249080238	12	512	3225.407974	LCD	5G	4.02184652	2	4	8.117500363	2022	330.035105	174.6703321	2.999521301	37.16918	4.54	Below Average	
3	3.401428613	4	256	4037.685111	LCD	4G	7.727446564	3	1	6.217241377	2025	282.784516	209.9001404	3.39787272	25.71991	3.46	Worst	
4	2.963987884	16	128	4999.379974	AMOLED	4G	8.577387451	6	3	5.226814612	2025	1077.01473	693.4698785	3.987230302	27.84983	6.15	Good	
5	2.697316968	12	512	3915.947778	LCD	5G	7.152696574	4	3	5.12924975	2020	1289.5239	1149.132286	3.719548438	33.04834	5.84	Below Average	
6	1.812037281	16	256	4722.475406	OLED	5G	6.585713988	8	2	8.780657861	2019	899.440819	407.51075	3.728134955	36.03306	6.69	Good	
7	1.811989041	6	128	4182.280828	AMOLED	4G	7.969807703	3	3	4.164130298	2022	227.132169	191.1931281	3.005760006	34.08092	2.86	Worst	
8	1.616167224	6	256	4017.850113	LCD	4G	9.609628486	2	3	7.212626418	2021	874.143782	509.5367273	3.099691941	30.75796	2.84	Worst	
9	3.232352292	6	512	4223.836275	LCD	5G	4.711982072	6	3	9.659106353	2016	1065.71599	827.3644292	2.597308728	35.83524	5.42	Below Average	
10	2.702230023	4	64	5609.049091	OLED	5G	9.014650754	4	3	4.207086387	2024	1017.70239	568.9527525	3.386152218	31.73914	4.81	Below Average	
11	2.916145156	6	128	3740.354145	AMOLED	5G	6.488998003	5	4	4.28801029	2017	502.732143	203.0738349	3.465489723	15.17379	5.07	Below Average	
12	1.541168989	4	64	5980.795822	LCD	5G	8.832977892	5	4	8.31770721	2025	471.546327	241.770237	2.572190608	27.84513	4.85	Below Average	
13	3.439819704	12	128	5715.218504	AMOLED	4G	7.282505189	2	4	8.093395216	2018	456.147637	185.097442	4.041814228	18.32444	5.11	Below Average	
14	3.164885282	4	512	3494.295989	LCD	5G	9.098990641	6	3	9.446483865	2023	368.475431	153.8735218	3.693113359	21.33371	5.67	Below Average	
15	1.924678221	8	128	4554.762292	LCD	5G	7.564715723	3	2	5.257494634	2024	1399.98979	833.3755269	4.463742015	32.35346	3.85	Worst	
16	1.863649934	16	128	5699.670235	AMOLED	4G	8.238702694	7	2	8.184675126	2015	243.151735	177.5578679	2.624170866	24.54756	4.82	Below Average	
17	1.86680902	12	256	3760.23572	AMOLED	5G	7.092871534	7	2	9.024458519	2025	589.515097	468.31821	2.922655616	19.69699	5.76	Below Average	
18	2.108484486	12	128	3554.482913	AMOLED	4G	6.443689128	7	3	6.900673396	2018	209.617046	141.3544124	3.688943798	24.23291	4.66	Below Average	
19	2.549512863	8	256	4419.973777	LCD	5G	6.606891148	8	2	9.863871966	2015	187.826555	76.60117781	2.642633558	33.34007	4.31	Below Average	
20	2.363890037	6	64	3736.55336	LCD	5G	9.421006783	6	3	5.382984465	2016	687.693409	606.6819184	3.703351861	36.83296	3.61	Worst	
21	2.08245828	6	128	5261.565939	LCD	4G	6.277522992	2	2	5.389234681	2023	941.618159	483.6161461	4.636966712	16.33401	3.61	Worst	
22	2.723705789	4	128	4582.749163	LCD	4G	5.632886526	4	3	8.92305004	2017	698.138271	610.727462	3.646430976	38.86424	2.78	Worst	
23	1.778987721	4	128	3937.140141	LCD	4G	8.493933721	5	4	6.416278179	2024	1394.13403	737.4744448	3.028186336	32.42733	3.88	Worst	
24	2.084289297	12	64	4475.090804	AMOLED	5G	9.981554479	5	3	9.559252026	2017	335.483678	184.2070846	4.436676953	20.84116	5.81	Below Average	
25	2.232723687	6	512	4753.00852	LCD	4G	6.621571131	4	2	7.203294931	2023	541.513901	385.1107307	3.679376923	33.31929	4.5	Below Average	
26	2.412139968	4	128	5343.21298	LCD	5G	4.205343943	5	2	9.190914717	2024	1472.5581	1242.038588	4.340473064	38.09708	4.7	Below Average	
27	3.070351923	8	128	4718.362067	AMOLED	5G	7.611393324	3	3	7.351819225	2023	625.681047	434.2068341	3.731848848	32.89859	4.93	Below Average	
28	1.899347564	8	1024	4818.436109	OLED	5G	4.371279368	6	1	4.241651632	2023	299.2646	158.9571239	3.363764397	16.9144	6.8	Good	
29	2.528468877	4	256	3247.218005	LCD	5G	8.633809158	8	2	8.721288274	2022	955.372778	810.9134724	3.251918075	21.831	5.47	Below Average	
30	2.684829138	6	256	3560.184666	AMOLED	4G	8.447403082	3	2	6.560093892	2023	172.261287	74.18416431	3.695305842	10.74805	4.1	Below Average	

2.2 Model Selection

In this project, several machine learning models are applied to analyse smartphone features and predict their obsolescence time. Since the problem involves both continuous and categorical target variables, the models are grouped into regression and classification approaches. Regression models are used to estimate the exact number of years a smartphone is expected to remain relevant, whereas classification models categorize devices into defined longevity tiers such as Best, Good, Below Average, and Worst. By experimenting with multiple algorithms in both categories, the study aims to evaluate how different learning strategies interpret hardware, software, and market-driven patterns present in the dataset. Among the classification models tested, the Support Vector Machine (SVM) achieved the highest accuracy of 82%, demonstrating its effectiveness in separating the obsolescence categories based on multidimensional smartphone attributes. The subsequent sections provide detailed descriptions of each model employed in the analysis.

2.2.1 Regression Models

Multiple Linear Regression

Multiple Linear Regression was the strongest-performing regression model in this study, achieving the highest accuracy in predicting the continuous obsolescence time of smartphones. Its effectiveness indicates that the relationship between smartphone features—such as processor speed, RAM, update support, and price—and their longevity can be approximated reasonably well through a linear combination of variables. The model captures the direction and magnitude of influence of each feature, making it highly interpretable and analytically insightful. Its performance suggests that much of the variance in obsolescence time can be explained by a linear structure embedded within the dataset.

Despite being one of the simplest models applied, its superior results demonstrate that increasing complexity does not always guarantee improved predictive capability. The synthetic dataset used in this project was designed with realistic yet structured feature relationships, and these structured relationships allowed the linearity assumption to hold effectively. As a result, Multiple Linear Regression outperformed more complex models such as Support Vector Regression and k-Nearest Neighbors Regression, making it the most reliable and computationally efficient option for this task.

k-Nearest Neighbors Regressor

The k-Nearest Neighbors (kNN) Regressor provides predictions by examining the most similar data points in the feature space. In the context of smartphone obsolescence, this means the model looks for devices with

similar specifications and market characteristics and averages their lifespans to estimate the target value. While this method can be powerful when local patterns dominate the dataset, it is highly sensitive to the distribution of features and requires careful scaling to ensure fair distance calculations. In this project, kNN delivered reasonable but not outstanding performance when compared to Multiple Linear Regression.

The lower performance of kNN relative to the best model may be attributed to the high dimensionality of the dataset. With many features influencing longevity, the distance between points becomes less meaningful—a phenomenon known as the "curse of dimensionality." Additionally, because kNN has no internal model and depends entirely on stored data, it is computationally expensive for large datasets such as ours. Although it provides insights into neighborhood-based prediction behaviour, it was not able to match the predictive strength of the linear approach used in the best-performing regression model.

Support Vector Regression

Support Vector Regression (SVR) seeks to fit an optimal function within a specified error margin, enabling it to model complex, non-linear relationships. While it is theoretically well-suited for datasets with intricate interactions, its performance in this project fell below that of Multiple Linear Regression. This suggests that although smartphone obsolescence patterns contain multi-factor influences, the structure of the synthetic dataset may favor more direct linear relationships, limiting the advantage of SVR's non-linear decision boundaries.

Additionally, SVR requires extensive tuning of its kernel parameters and regularization values to achieve optimal results. The high computational cost associated with training SVR on a large dataset may also limit its efficiency. Although it captured some non-linear patterns in the data, the incremental benefit did not surpass the simpler linear model. Therefore, SVR served as a strong comparative tool but did not emerge as the top regression performer in this study.

Random Forest Regressor

The Random Forest Regressor is an ensemble method that builds multiple decision trees and averages their outcomes to produce stable predictions. It is highly effective in capturing non-linear interactions and identifying feature importance. In this project, the Random Forest Regressor exhibited solid performance and demonstrated its ability to learn from diverse smartphone attributes, including hardware, software support, and market indicators. However, it did not outperform the simpler Multiple Linear Regression model.

Random Forest's slightly lower performance can be attributed to the synthetic dataset's strong linear ten-

dencies. While ensemble models typically excel in complex real-world settings, this dataset's structured design allowed the linear model to fit the underlying relationships more accurately. Although Random Forest did not provide the best regression accuracy, it still offered valuable interpretability through feature importance analysis and served as a robust comparative model.

2.2.2 Classification Models

Support Vector Machine Classifier

The Support Vector Machine (SVM) Classifier achieved the highest classification accuracy of 82% in predicting smartphone obsolescence categories. Its strength lies in its ability to construct an optimal separating hyperplane that maximizes the margin between different classes. This is particularly beneficial for this dataset, where the obsolescence categories exhibit overlapping yet distinct boundaries influenced by multiple interacting features. The SVM's capability to operate effectively in high-dimensional spaces allowed it to capture subtle patterns in hardware, software, and market attributes that significantly influence longevity.

The superior performance of SVM compared to other classification models highlights its robustness and ability to generalize well. While models like Decision Trees or kNN may struggle with noisy boundaries or high feature correlation, SVM maintains a consistent separation strategy, reducing misclassification. The use of appropriate kernel transformations further enhances its ability to model non-linear separations, making it the most reliable classifier for this project.

k-Nearest Neighbors Classifier

The k-Nearest Neighbors Classifier categorizes devices based on the majority class of their closest neighbors in feature space. While intuitive and easy to implement, its reliance on measuring distances between high-dimensional data points can reduce accuracy when the dataset becomes large and complex. In this project, kNN provided moderate performance but failed to match the precision and robustness demonstrated by SVM. This is likely due to the dataset's dimensionality and the presence of features with differing scales, which can distort distance calculations even after normalization.

Additionally, kNN is sensitive to noise, and smartphones with borderline specifications may be misclassified if their nearest neighbors belong to different longevity categories. Although the model captures local structure effectively, global classification boundaries are not well-represented. This limitation resulted in lower performance compared to SVM, making kNN a useful benchmark but not a leading classifier in this study.

Decision Tree Classifier

The Decision Tree Classifier builds a hierarchical structure where data is split based on feature thresholds to assign class labels. It performed reasonably well in the project and successfully captured non-linear relationships among features. However, its tendency to overfit, especially when the dataset is large and contains many correlated features, prevented it from outperforming SVM. While the tree structure is highly interpretable, it may create complex decision regions that do not generalize well.

Despite these challenges, Decision Trees still demonstrated meaningful classification capability. They quickly adapt to the most informative features and provide transparent decision rules. However, their instability—small changes in input data can cause large changes in the tree—made them less reliable than the more stable SVM classifier, which ultimately delivered the best predictive results.

Random Forest Classifier

The Random Forest Classifier aggregates multiple decision trees to reduce overfitting and improve classification stability. In this project, the model performed well and demonstrated its strength in handling non-linear relationships and mixed feature types. Random Forest identified important factors influencing obsolescence categories and provided strong interpretability through feature importance analysis. However, despite these advantages, it did not surpass the performance of SVM.

The ensemble nature of Random Forest allowed it to generalize better than single decision trees, but its classification boundaries were not as precise or well-defined as those created by SVM. The SVM's margin-maximization strategy proved more effective in handling the subtle boundaries between the four obsolescence categories. Therefore, although Random Forest was one of the stronger models evaluated, SVM remained the best-performing classifier.

2.3 Gradio Interface Implementation

To make the outcome of the machine learning model more accessible and user-friendly, a Gradio interface was developed as the final part of this project. Gradio is an interactive tool that allows machine learning systems to be deployed as simple web applications, enabling users to enter input values and instantly receive predictions. The primary purpose of integrating this interface was to create a practical and interactive environment where anyone can experiment with different smartphone specifications and observe the predicted obsolescence category without having to understand or run Python code.

The interface allows users to enter all the major smartphone features that influence obsolescence, such as processor capability, memory, storage, battery capacity, display type, connectivity features, camera quality, and software support details, along with market-oriented factors like brand strength, launch year, price, and customer rating. These inputs are presented in a clean and organized layout, making it easy for users to adjust values and explore how changes in specifications affect the predicted longevity of the device. This design ensures that the interface is intuitive and can be easily operated by non-technical users.

Once the user enters the details, the interface processes the information internally and uses the trained machine learning model to determine the most appropriate obsolescence category for the smartphone. The model evaluates the input features and classifies the device into one of the predefined categories based on its expected longevity. The predicted class is then displayed clearly on the screen, providing immediate feedback to the user. This makes the interface highly interactive and allows users to experiment with various combinations of specifications to understand what factors contribute most to the overall lifespan of a smartphone.

The Gradio interface also enhances the practical value of the project by demonstrating how machine learning models can be deployed in real-world scenarios. Instead of remaining confined to a programming environment, the predictive model becomes accessible like a simple online tool. This makes it useful not only for demonstration and academic purposes but also for potential applications among consumers, retailers, and device evaluators who wish to assess how long a smartphone is likely to remain relevant.

Overall, the integration of Gradio transforms the machine learning model into an interactive and easy-to-use application that bridges the gap between complex technical development and everyday usability. By providing instant predictions based on user input, the interface adds a practical dimension to the project and showcases how predictive analytics can be translated into meaningful decision-support tools for smartphone assessment.

3 Implementation

4 Implementation

The implementation of this project involves a systematic series of steps beginning with the creation of a synthetic dataset that simulates real-world smartphone specifications, software support indicators, and market-driven attributes. Once the dataset was generated, several preprocessing operations were applied to encode categorical features, normalize variables, and prepare the data for machine learning. After the data was cleaned and transformed, various regression and classification algorithms were implemented to predict the obsolescence time and the corresponding longevity category of smartphones. Evaluation metrics appropriate to each type of model were used to analyze performance. Finally, a Gradio interface was created to convert the trained model into an accessible, user-friendly tool. The complete implementation workflow is described in the following subsections.

4.1 Dataset Creation

The dataset used in this project was synthetically generated using NumPy and Pandas to simulate 80,000 smartphone entries. Each smartphone was assigned realistic hardware specifications such as processor speed, RAM, storage, and battery capacity. Additional categorical attributes such as display technology and connectivity type were generated based on probability distributions reflecting real-world market trends. Software support indicators—including update support duration and number of security updates per year—were also incorporated.

Market-oriented features such as brand score, release year, pricing, resale value, customer rating, and depreciation rate were generated to reflect consumer perception and device lifecycle patterns. Using these features, a weighted formula was constructed to compute the *Obsolescence_Time_Years*, after which it was normalized to a 1–10 scale and categorized into four classes: Best, Good, Below Average, and Worst. This created a comprehensive dataset suitable for regression and classification tasks.

4.2 Data Preprocessing

Before training the models, preprocessing steps were applied to ensure the dataset was clean and suitable for machine learning. Categorical variables such as display technology and connectivity type were encoded using Label Encoding. The target class labels were mapped to numerical values using a custom dictionary: Best $\rightarrow$ 3, Good $\rightarrow$ 2, Below Average $\rightarrow$ 1, and Worst $\rightarrow$ 0. Continuous variables were scaled using standardization

to ensure uniform contribution across features.

A correlation heatmap was generated to understand feature relationships and identify the influence of hardware, software, and market attributes on the obsolescence outcome. The dataset was then divided into training and testing sets to evaluate model generalizability.

4.3 Regression Implementation

Regression models were trained to predict the continuous variable *Obsolescence_Time_Years*. To evaluate the performance of these models, two standard metrics were used: Mean Squared Error (MSE) and the Coefficient of Determination (R^2 Score).

Mean Squared Error

Mean Squared Error measures how close the model's predictions are to the actual values. It penalizes large errors more heavily and is defined as:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

where y_i is the actual obsolescence time, $\hat{y}_i$ is the predicted value, and n is the total number of samples.

R^2 Score

The R^2 score measures the proportion of variance in the dependent variable that can be explained by the model. It is calculated using:

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

where $\bar{y}$ is the mean of the actual values. An R^2 value closer to 1 indicates that the model captures more variability in the data.

4.4 Classification Implementation

Classification models were implemented to categorize smartphones into four groups based on their obsolescence time. The performance of these models was assessed using two metrics: Accuracy and the Confusion Matrix.

Accuracy

Accuracy represents the proportion of correctly classified samples and is defined as:

$$Accuracy = \frac{Number\ of\ Correct\ Predictions}{Total\ Number\ of\ Predictions}$$

Higher accuracy indicates better model performance. In this project, Support Vector Machine (SVM) achieved the highest accuracy among the models tested.

Confusion Matrix

The confusion matrix provides a detailed breakdown of predictions across classes. For a four-class classification problem, the confusion matrix is represented as:

$$CM = \begin{bmatrix} TP_1 & FP_1 & FN_1 & \dots \\ FP_2 & TP_2 & FN_2 & \dots \\ \dots & \dots & \dots & \dots \\ FN_4 & FP_4 & \dots & TP_4 \end{bmatrix}$$

where,

$TP = True\ Positive$

$FP = False\ Positive$

$TN = True\ Negative$

$FN = False\ Negative$

This matrix helps identify misclassification patterns across categories.

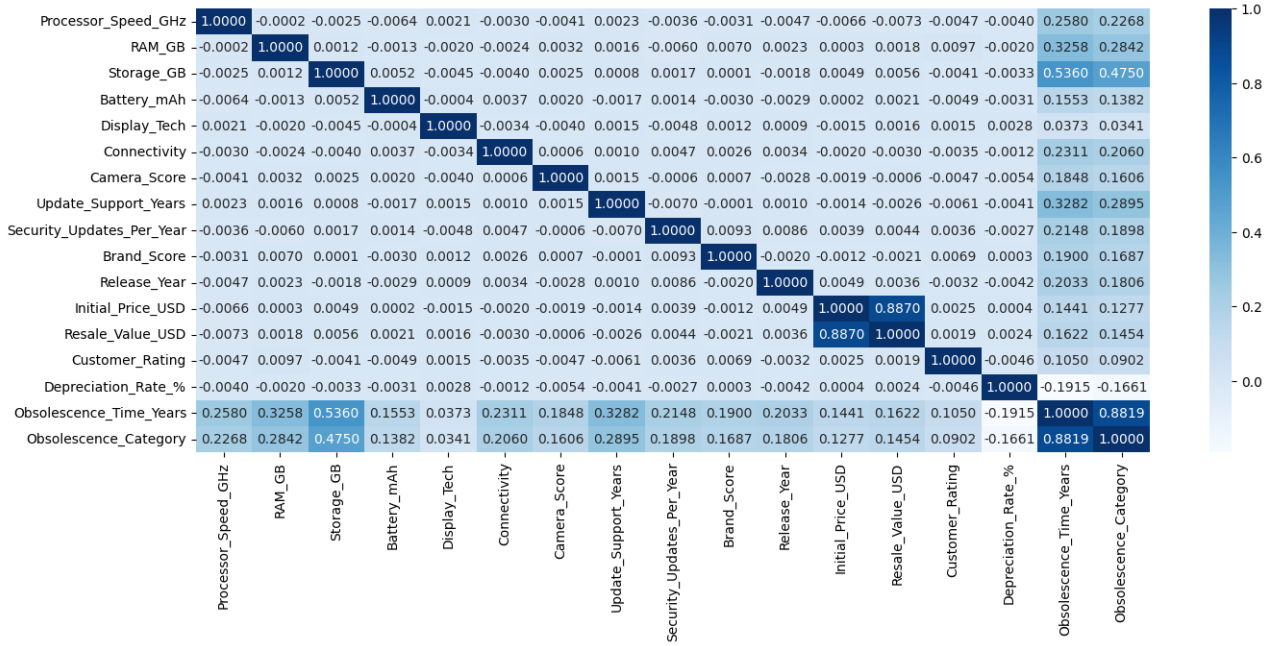
4.5 Gradio Interface Implementation

To ensure that the final prediction system could be used easily by non-technical users, a Gradio interface was developed. This interface allows users to input key smartphone attributes such as processor speed, RAM, storage capacity, display type, connectivity, camera score, software support details, brand score, release year, price, and customer rating. Once the user enters the values, the system processes the inputs, applies the same preprocessing steps used during model training, and then generates the predicted obsolescence category.

The interface provides instant results, making it a practical and interactive tool for exploring how different smartphone specifications influence longevity. By deploying the machine learning model through Gradio, the project bridges the gap between technical implementation and real-world usability, demonstrating the practical application of predictive analytics in evaluating smartphone lifespan.

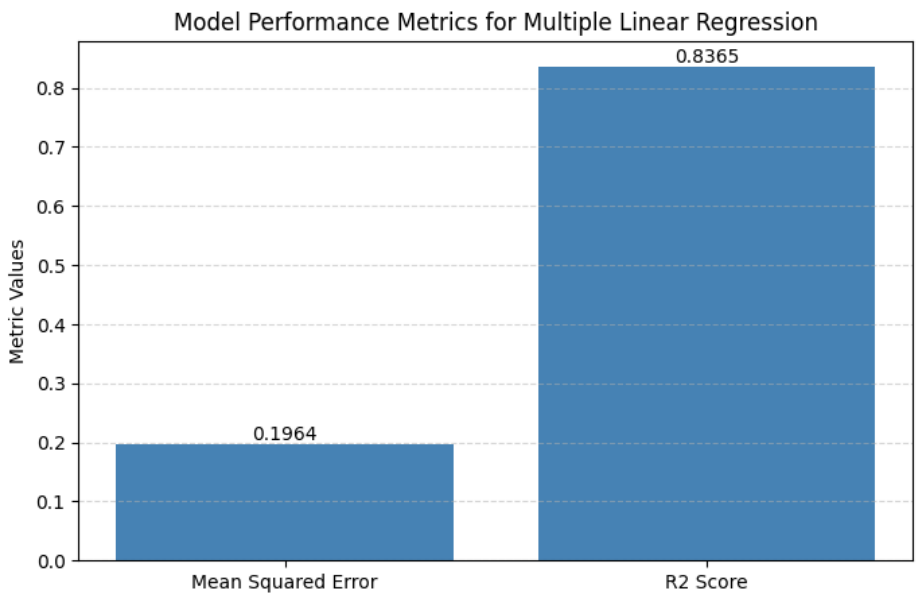
5 Result Analysis

5.1 Correlation

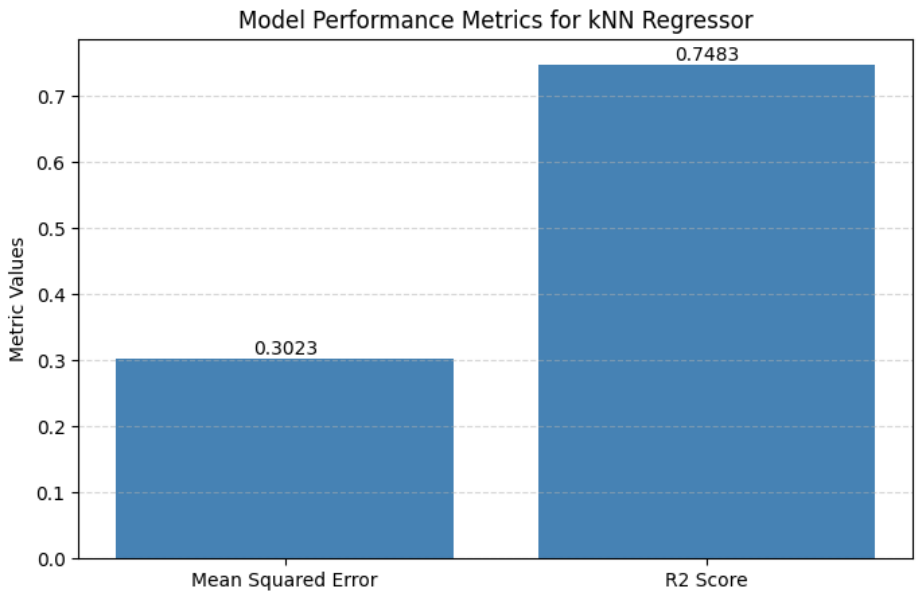


5.2 Regression Results

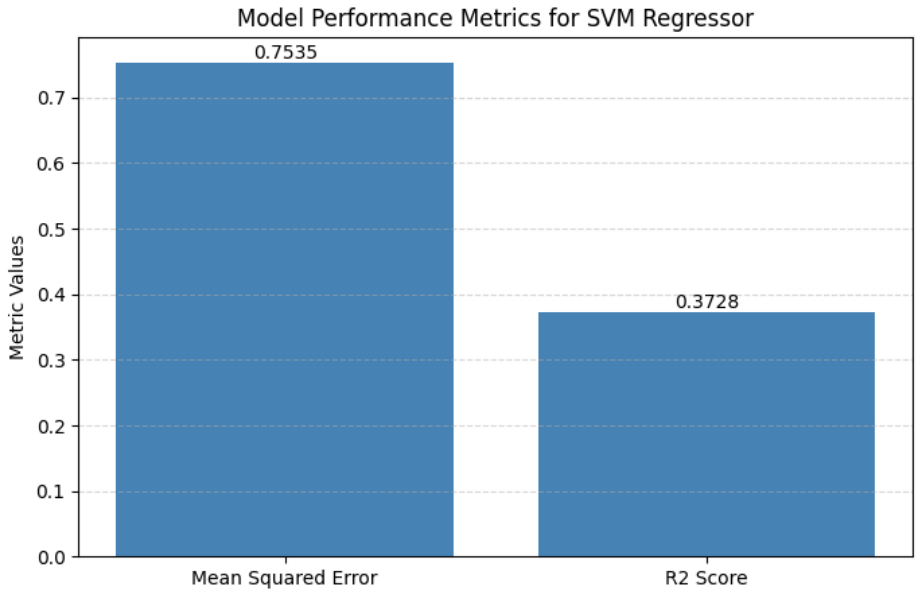
Multiple Linear Regression



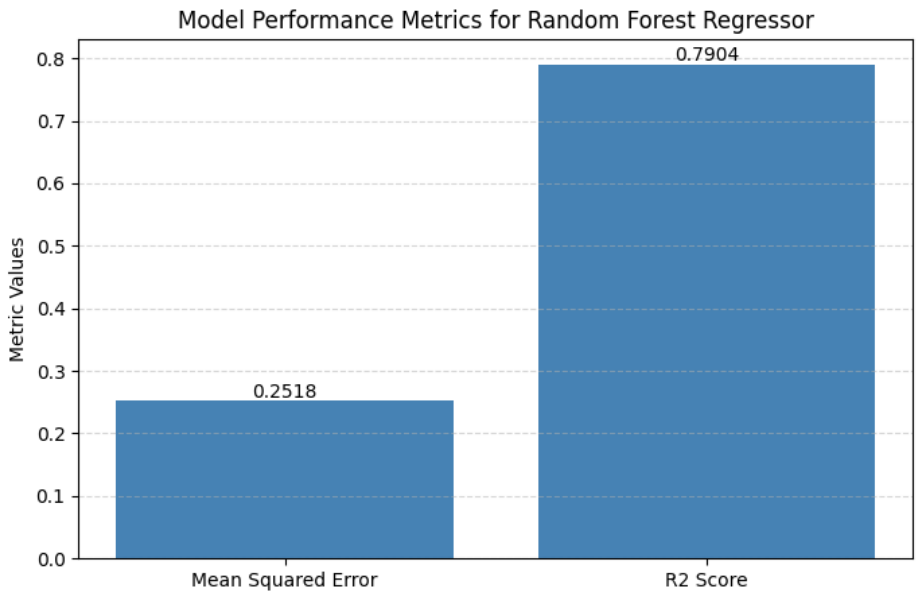
kNN Regressor



SVM Regressor

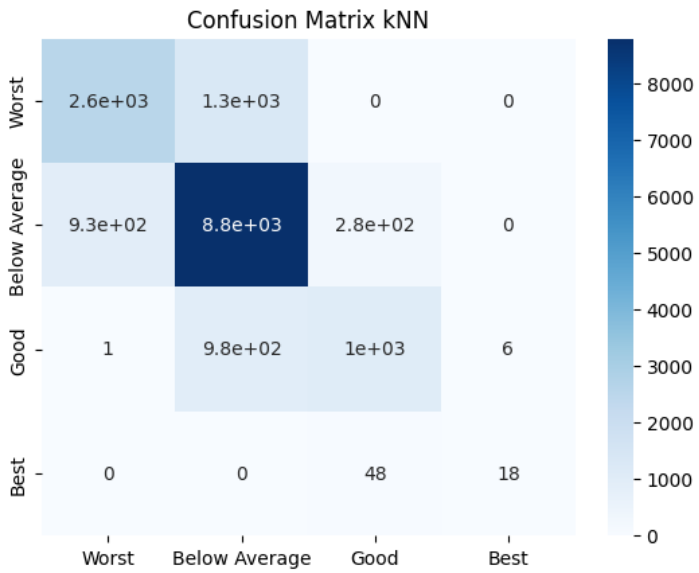


Random Forest Regressor



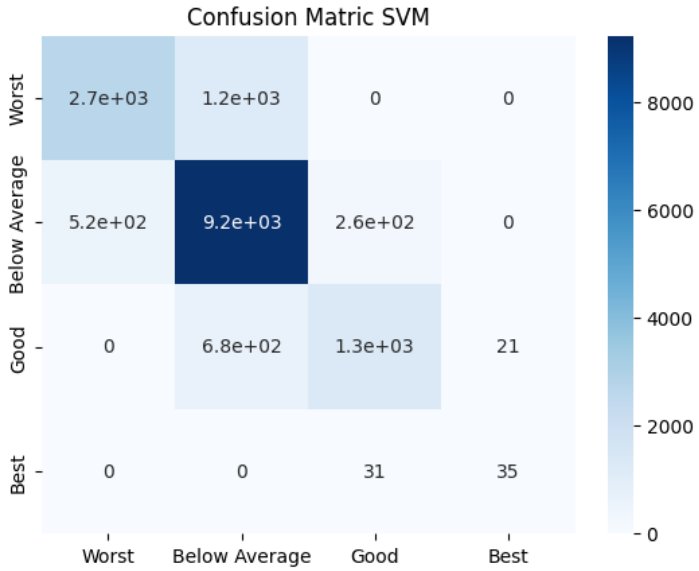
5.3 Classification Results

kNN Classifier



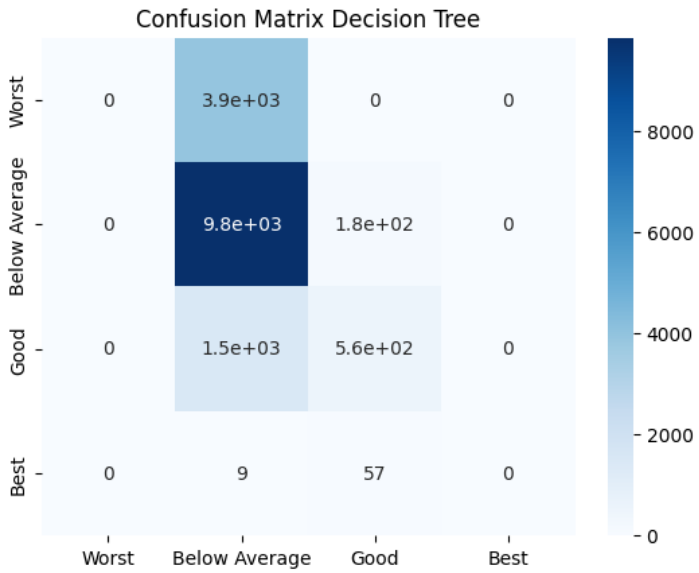
Accuracy = 0.78

SVM Classifier



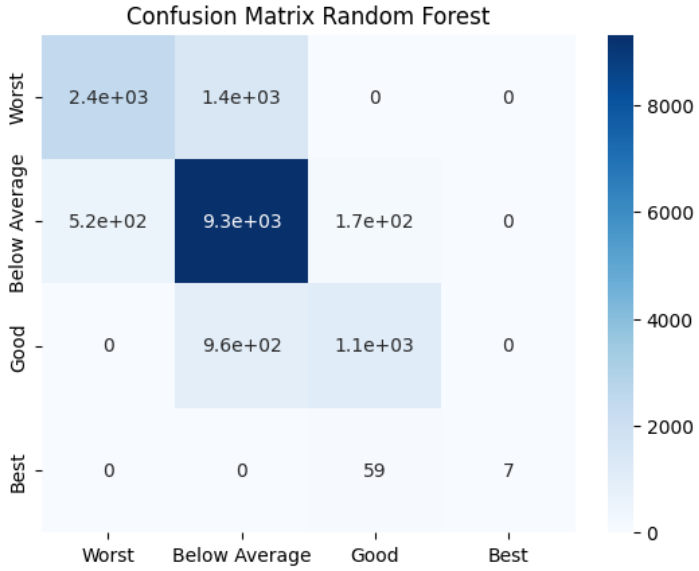
Accuracy = 0.83

Decision Tree Classifier



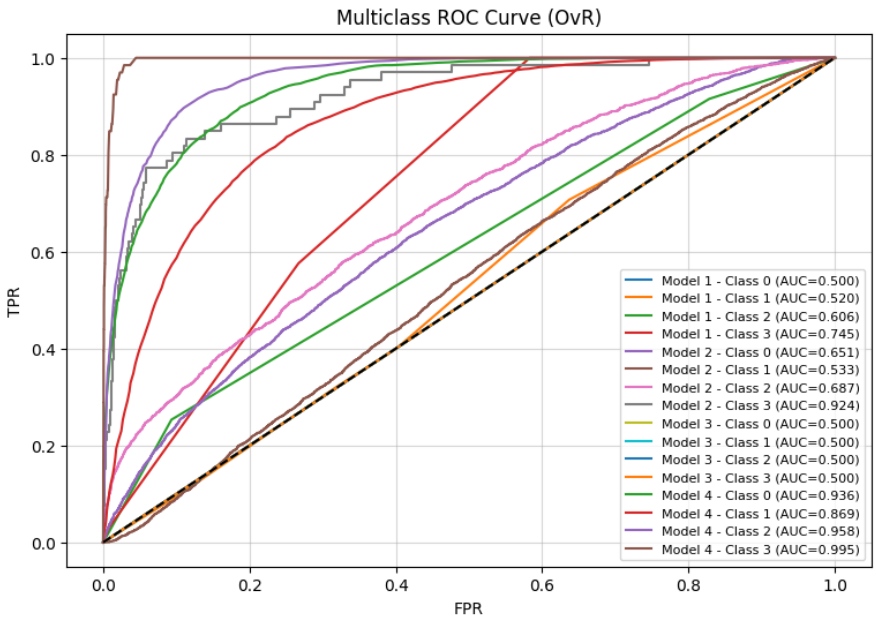
Accuracy = 0.65

Random Forest Classifier



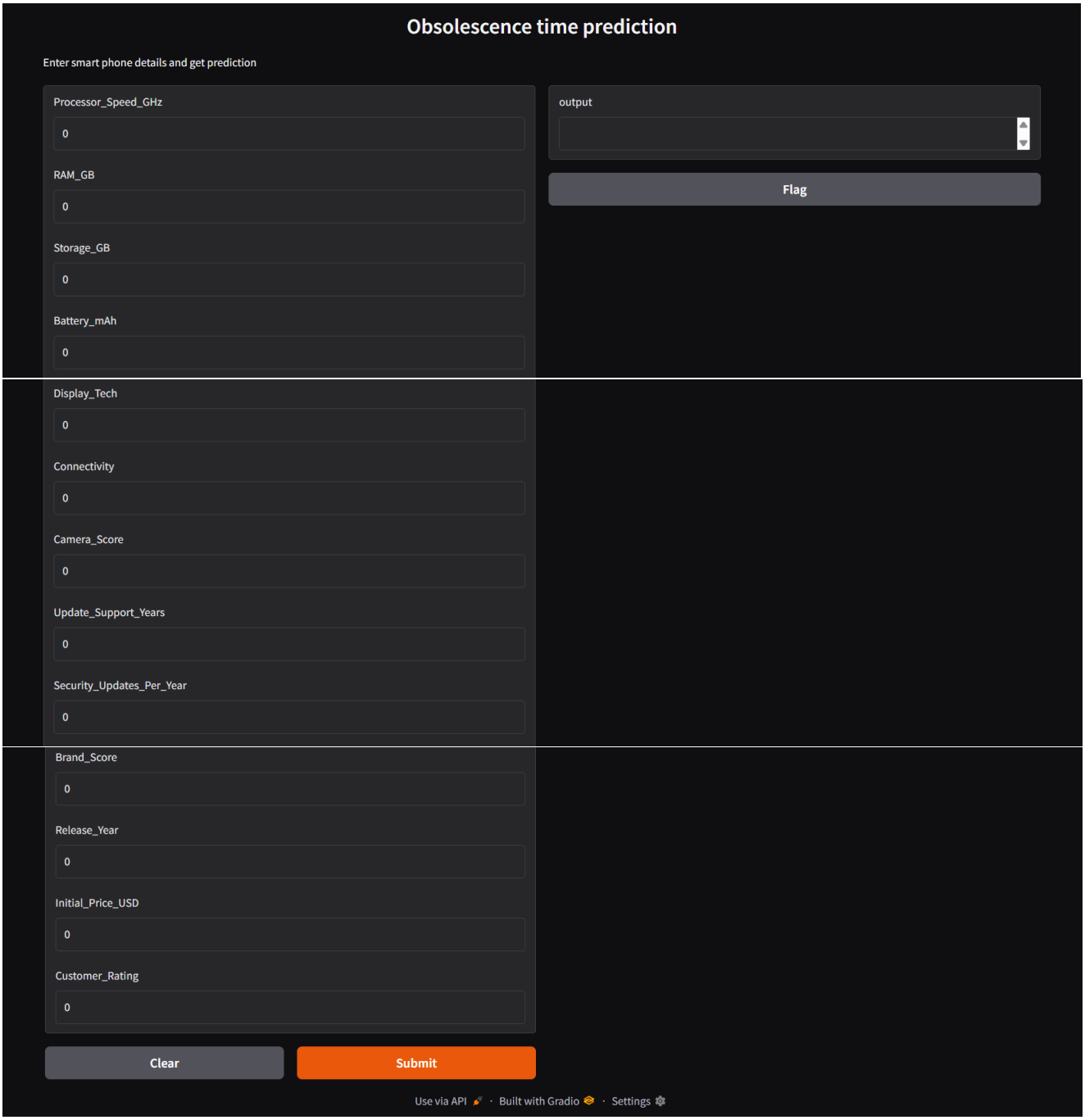
Accuracy = 0.0.81

Comparative Result

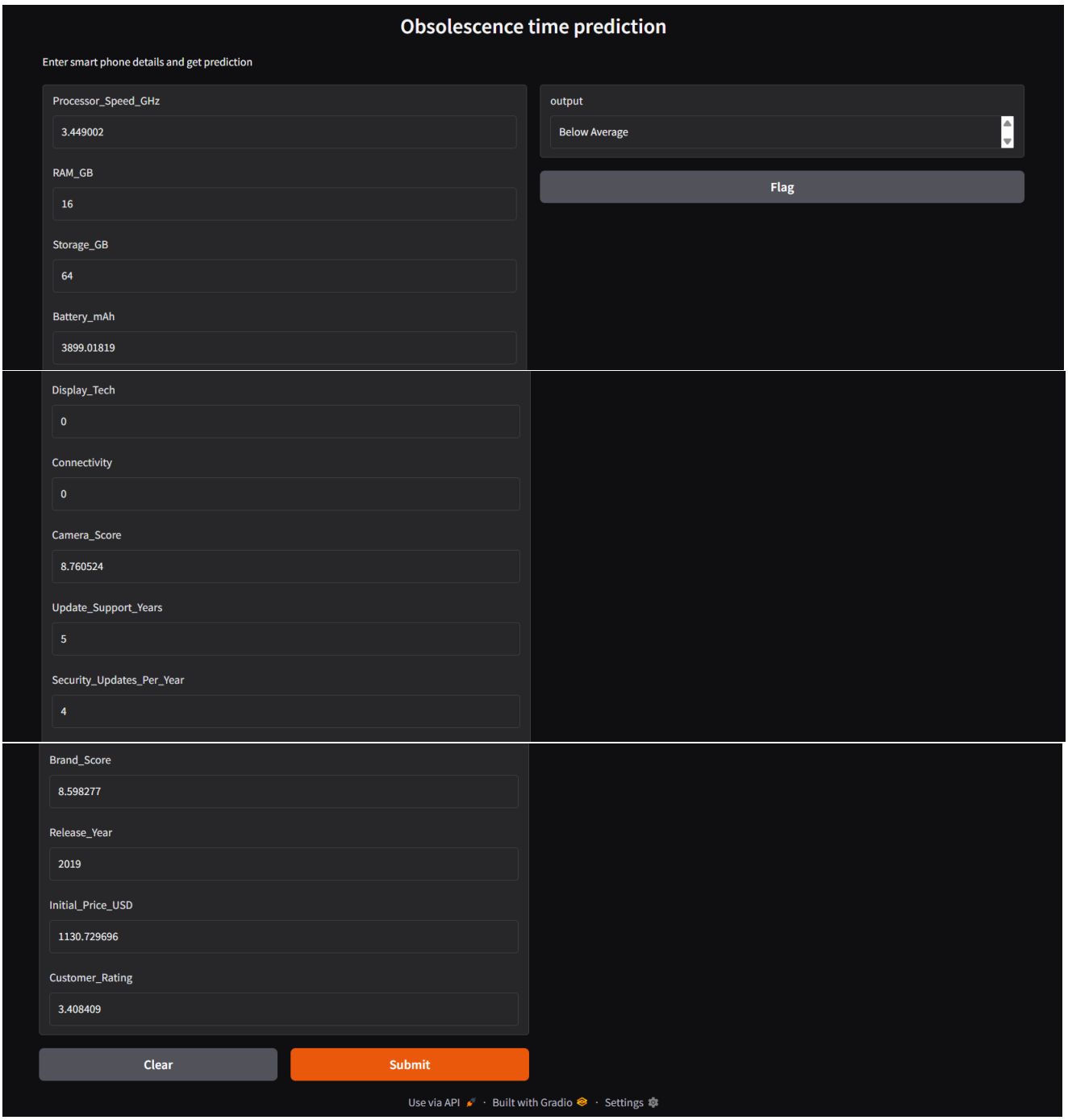


5.4 Gradio Interface

Before giving inputs



After giving inputs



6 Conclusion

This project successfully demonstrates that the obsolescence time of a smartphone can be effectively predicted using machine learning techniques by leveraging a combination of hardware specifications, software support indicators, and market-based attributes. By generating a large and statistically realistic synthetic dataset, the study was able to model diverse smartphone configurations and simulate how various factors contribute to a device's long-term usability. The comprehensive preprocessing pipeline—consisting of categorical encoding, normalization, and correlation analysis—ensured that the data was well-structured and suitable for training robust models.

The implementation of both regression and classification algorithms provided two complementary perspectives on smartphone longevity. Regression models offered continuous predictions of usable lifespan, while classification models enabled the grouping of devices into clear categories such as Best, Good, Below Average, and Worst. Among the regression models, Multiple Linear Regression performed most reliably, showing that the dataset's underlying relationships were predominantly linear in nature. For classification, the Support Vector Machine (SVM) emerged as the most effective model, achieving the highest accuracy of 83% and demonstrating strong capability in handling complex, multidimensional feature interactions.

In addition to the analytical components, the development of a Gradio interface played a crucial role in enhancing the practical relevance of the work. This interface allowed the trained model to be deployed as an interactive web application, enabling users to input smartphone characteristics and receive real-time predictions without requiring any programming knowledge. This bridge between technical implementation and real-world accessibility highlights how predictive models can support informed decision-making for consumers, researchers, and manufacturers alike.

Overall, the project validates the feasibility of predicting smartphone longevity using machine learning and provides a scalable foundation for more advanced future studies. With further refinement—such as incorporating real-world datasets, modelling software aging more precisely, or integrating time-series behaviour—this approach can contribute to sustainable device usage, improved consumer awareness, and more strategic planning within the smartphone industry